

RESEARCH ARTICLE

Lattice Slot Waveguide for Terahertz Microfluidics Biomedical Trace Analysis

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Received: 5 November 2025 | **Revised:** 25 January 2026 | **Accepted:** 4 February 2026

Keywords: biosensors | lattice resonance | metasurface | microfluidics | terahertz spectroscopy

ABSTRACT

Electromagnetic metasurface integrated microfluidic chips enable a real-time, label-free platform for terahertz trace analysis of volume-limited biomedical samples with suppressed water absorption noise. However, conventional metal-insulator-metal (MIM) metasurface resonators exhibit inherently limited Q-factor and sensitivity due to radiative leakage through open side boundaries. Here, a lattice slot waveguide based on MIM configuration is designed to effectively confine energy within the microfluidic channel and mitigate radiative loss. This trapped mode achieves enhanced sensitivity and Q-factor through synergistic excitation of surface lattice resonance and guided mode resonance under propagation constant matching conditions. Leveraging this platform, an anisotropic detection strategy incorporating a patterned lattice structure is devised to achieve simultaneous polarization multiplexed responses, exhibiting a figure of merit of 135 in both polarizations. Experimental validation demonstrates a limit of detection of 625 pmol mL⁻¹ and a Q-factor of 189 for this polarization multiplexing microfluidic platform. This work offers a unique avenue for enhanced accuracy and efficiency in terahertz biomedical trace analyzing via multidimensional sensing capabilities.

1 | Introduction

Terahertz (THz) technology offers significant potential for next-generation communication, imaging, and detection systems, leveraging its unique spectral properties such as low photon energy and exceptional material penetration capabilities [1–3]. Crucially, the intrinsic vibrational frequencies of biomacromolecules reside within the terahertz band, enabling promising biomedical applications including molecular fingerprint identification and trace analysis [4–7]. However, these biomarkers typically exist in aqueous environments, where water molecules exhibit orders-of-magnitude stronger THz absorption than target solutes. This interference drastically reduces the signal-to-noise ratio (SNR), presenting a fundamental limitation for THz-based

biosensing [8, 9]. A common mitigation strategy involves sample dehydration, where liquid samples are transformed into thin films on sensor surfaces via controlled heating [10]. Numerous resonant absorbers with enhanced sensing performance were subsequently proposed [11–13]. These methods perform effectively in the absence of water interference; however, they still encounter several challenges. On the one hand, the heating process during drying may degrade the molecular structure of target biomacromolecules, compromising sensing accuracy [14, 15]. On the other hand, for analytes of unknown concentration, unpredictable film thickness post-drying can perturb resonant frequencies [16, 17]. Consequently, a direct liquid state detection methodology is essential yet remains challenging for terahertz biomedical sensing.

Recently, lab-on-chip systems integrating microfluidics with electromagnetic metasurfaces have emerged as a promising platform for THz biosensing. Precise microliquid manipulation enhances analysis speed while minimizing sample consumption [18–20]. Given the micron-scale wavelengths of THz radiation, the dimensions of microfluidic channels are inherently compatible, offering a pathway to reduce water absorption noise and enable high sensitivity aqueous detection. The integration of metasurfaces into microfluidics significantly enhances sensing sensitivity, as it allows minute refractive index changes to be transduced into measurable frequency shifts [21–24]. Transmission type metasurfaces have been widely studied due to their operational simplicity and flexible optical configuration [25]. However, they suffer from refractive index (RI) mismatch, while incorporating compensation layers inevitably disrupts light-matter interactions [26]. This has shifted focus toward reflective metasurfaces, typically based on MIM resonators [18]. Achieving high performance in reflective MIM resonators requires precise control over resonator height, which is a parameter highly sensitive to fabrication tolerances, to minimize radiation loss from the lateral sides. Combined with intrinsic metallic Ohmic losses, these constraints limit the quality factors (Q-factors) of microfluidic sensors based on reflective metasurfaces to relatively low values (~ 10) [18, 23]. While versatile metasurface designs have recently been integrated into microfluidics [22, 23, 27–31], achieving higher quality factors remains challenging. Although ultra-high Q strategies such as quasi-bound states in the continuum (qBIC) in silicon metasurfaces exist [32–36], their 3D or enclosed structures are often incompatible with thorough liquid cleaning and device reusability. In contrast, metasurface designs suitable for practical microfluidic integration typically exhibit limited Q-factors. Therefore, attaining higher performance requires resonant modes capable of minimizing lateral radiation loss while maintaining insensitivity to microfluidic channel height.

Surface lattice resonance (SLR) is a collective resonance from surface plasmons with relatively high Q-factor [37]. Its resonance wavelength is primarily determined by the background refractive index and lattice period [37–41], rendering its performance insensitive to unit cell structural fabrication tolerances [39]. Proper optimization of period and nanostructure dimensions can enhance SLR Q-factors to the order of 10^3 in transmissive configurations [42]. Despite this advantage, SLR sensitivity is often limited by weak field confinement at the substrate surface, typically addressed by pre-measurement sample drying [43]. Different from typical THz fibers that support transmission wave modes [44, 45], guided mode resonance (GMR) provides an alternative route to ultra-high Q-factors (up to 10^5) [46–48]. Integration with grating slot waveguides (GSWs) concentrates fields and analytes within the slot layer, enabling simultaneous high Q-factor and sensitivity [49]. Nevertheless, GMR performance remains critically dependent on slot waveguide height and polarization direction. Furthermore, the inherent slot structure significantly complicates cleaning procedures for microfluidic chips, further limiting its practical versatility.

To advance sensing capabilities, multi-dimensional detecting strategies have been put forward to amplify THz absorption features for fingerprint recognition [50–52], such as multiplexing in space [39, 40], frequency [27], polarization [53, 54], angle [55, 56], chirality [57, 58] and phase [59]. Such approaches enable concur-

rent analyte interrogation from multiple perspectives, enhancing sensing efficiency, precision, capability, and robustness. However, realizing a polarization multiplexing microfluidic sensor with overall high performance remains unexplored, primarily due to inherent trade-offs with spatial, frequency, or other sensing parameters.

Here, we propose a methodology for enhancing the aqueous sensing limit based on a lab-on-a-chip lattice slot waveguide (LSW) integrated microfluidic structure. We investigate a hybrid resonance combining SLR and GMR, designed to decouple radiation loss from channel height constraints inherent in conventional MIM resonators. Furthermore, we explore a novel polarization multiplexing sensing strategy empowered by a lattice weaving operation applied to the LSW. This work demonstrates a terahertz microfluidic meta-sensor with anisotropic sensing capabilities, achieving a maximum figure of merit (FOM) of 135 in both polarized modes. This performance would pave the way for advanced biosensing, trace analysis, and molecular identification via combined polarization and frequency multiplexing.

2 | Results

2.1 | Excitation of Lattice Slot Waveguide

We designed a reflective microfluidics biosensor leveraging collective resonances with LSW architecture. As schematically illustrated in Figure 1A, the device is a reflective metasurface integrated microfluidics based on MIM configuration. Crucially, the hydrated analyte serves as the insulator layer within this structure. Liquid samples are circulated through integrated inlet/outlet ports, enabling reusable and efficient operation. Meta-atoms on the surface of microfluidics govern the collective frequency response along the lattice direction. Therefore, this design facilitates independent sensing operations for distinct polarizations.

Our approach begins with the LSW design, engineered to support a hybrid resonance combining SLR and GMR for enhanced aqueous sensing. Figure 1B details the unit cell of configuration, where SLR meta-atoms are fabricated on a polyimide (PI) cap ($\epsilon_r = 3.5 + 0.05i$) [27, 28] and a metallic reflector is deposited on a silicon holder. A slot waveguide (SW) sustaining GMR is sandwiched between metallic layers, subsequently functioning as the microfluidic channel. Numerical simulations (Figure 1C) reveal two resonances in the LSW under Y-polarized THz illumination. Upon coupling with GMR, the SLR resonance exhibits a significantly narrower full-width-at-half-maximum (FWHM) compared to the localized surface plasmon resonance (LSPR) mode within the LSW. This narrowed SLR linewidth originates from collective resonance between dipole modes and Rayleigh anomalies (RAs), where engineered periodic perturbations are introduced into the dipole array to amplify mutual coupling, thereby achieving spectral sharpening. Following Bloch wave theory and Rayleigh anomaly (RA), the SLR resonant wavelength (λ_{SLR}) can be approximated as Equation 1 [38, 42].

$$\lambda_{SLR} = n_{eff}P \quad (1)$$

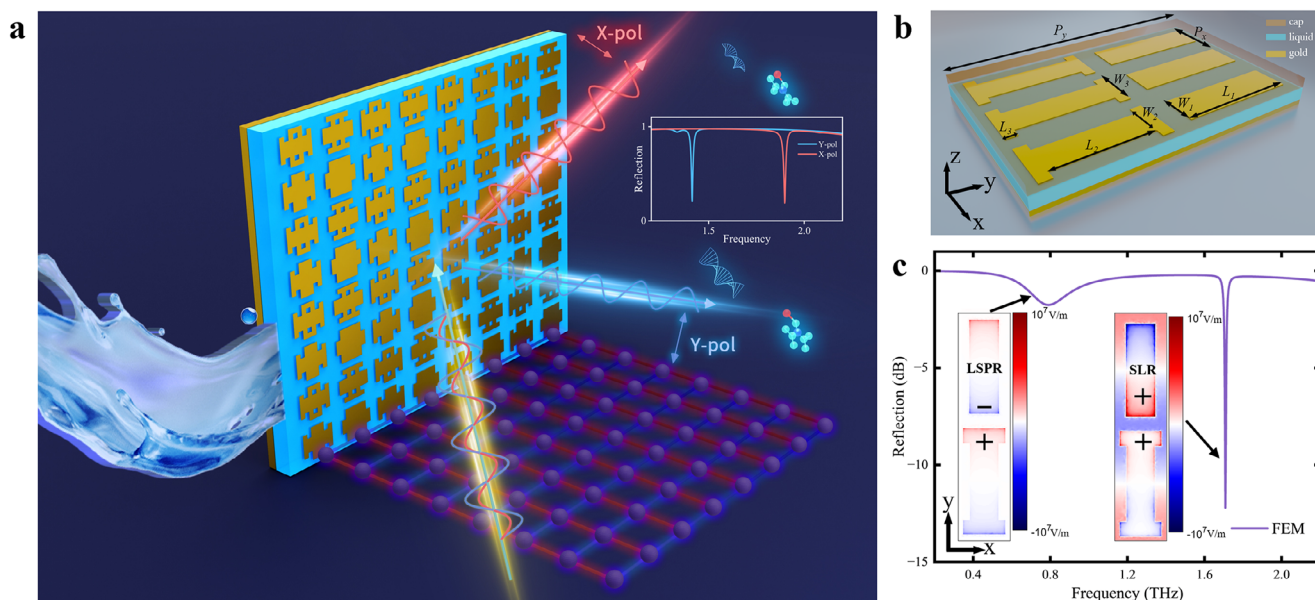


FIGURE 1 | Schematic of LSW. (A) Schematic diagram of the LSW sensor with polarization multiplexing capability. (B) unit cell of LSW metasurface. Here $P_y = 110 \mu\text{m}$, $P_x = 25 \mu\text{m}$, $L_1 = 45 \mu\text{m}$, $L_2 = 37 \mu\text{m}$, $L_3 = 7 \mu\text{m}$, $W_1 = 14 \mu\text{m}$, $W_2 = 13.5 \mu\text{m}$, $W_3 = 20 \mu\text{m}$. (C) Numerical (FEM) calculations of the reflection of the microfluidic for y-polarized wave, where LSPR and SLR are excited simultaneously in LSW. Inset: The simulated E-field magnitude in z-direction for the unit cell of LSPR and SLR.

where n_{eff} is the effective RI of the background medium, P is the lattice period in polarization direction. Meanwhile, GMR arises from the resonant coupling between incident light and guided modes within the waveguide structure, mediated by a grating coupler. This resonance is determined by the Bragg condition, satisfying the wavevector matching requirement between the diffracted wave and the waveguide mode [49]:

$$n_{\text{eff}} \pm n_{\text{air}} \cdot \sin \theta = m \frac{\lambda_{\text{GMR}}}{\Lambda} \quad (2)$$

where θ is incident angle, m is the diffraction order, Λ is the period of gratings. According to coupled mode theory (CMT), a perfect mode coupling occurs under propagation constant matching ($\Delta\beta = \beta_{\text{SLR}} - \beta_{\text{GMR}} = 0$). In our LSW design, this condition corresponds to matching the lattice period and grating period ($P = \Lambda$) for the fundamental diffraction order ($m = 1$). Finite Element Method (FEM) simulations in Figure 1C reveal that the SLR excites a distinct grating like electric field distribution within the SW, analogous to conventional GMR in GSW known for exceptional Q-factors. Critically, Figure 1C demonstrates a collective resonance mechanism fundamentally distinct from typical LSPR. Unlike LSPR where adjacent meta-atoms exhibit in-phase E-fields, the SLR manifests out-of-phase field distributions characteristic of diffractive coupling. More significantly, we observe an average 10^2 times enhancement of the electric field magnitude within the metallicly uncovered SW region compared to LSPR modes. This substantial enhancement stems from the optimized propagation constant matching between the SLR and the fundamental mode of the SW, confirming that the SLR-GMR hybridization provides a viable pathway to dramatically enhance field confinement and spatial overlap with analytes in the microfluidic channel. This synergy directly translates to superior sensing performance.

Thus, our LSW design demonstrably exhibits simultaneous excitation of localized surface plasma and guided mode characteristics. On the one hand, the SLR establishes a standing wave mode on the upper metallic layer of the microfluidic channel, significantly minimizing lateral radiation losses through diffractive confinement. On the other hand, the confined SLR field directly matches with the GMR mode within the microfluidic channel, maximizing the light matter interaction. Consequently, the electromagnetic fields and target analytes experience substantial spatial overlapping within the sensing region. This synergy excites a hybridized resonance mode within the side-open MIM resonator, delivering simultaneously high Q-factor and sensitivity. This enhanced performance overcomes fundamental limitations of conventional aqueous THz biosensors.

To verify resonance hybridization, we performed detailed mode analysis of the LSW. Figure 2A compares lateral view of LSPR and SLR modes under FEM simulation. The LSPR propagation constant depends solely on dielectric properties at the metal interface, preventing coupling into the SW and confining it to surface resonance, which is a fundamental limitation of MIM resonators in microfluidics and imposes strict restriction on channel height practically. In contrast, SLR exhibits significantly stronger E-field penetration into the SW. As theoretical predicted in Equations 1 and 2, propagation constant matching ($\Delta\beta = 0$) is achieved when $\Lambda = P$. Mode analysis confirms the SLR on the upper metal layer excites a grating-like E-field distribution, establishing conditions for GMR excitation within the microfluidics [49]. Unlike conventional surface resonances, this hybrid mode couples SLR with matched GMR, distributing energy horizontally as a lattice confined TM mode. This creates a unique resonance with minimal radiation loss. To quantitatively evaluate the field enhancement capability, we defined a factor of Average Field Enhancement Ratio (AFER), which points to the volume averaged electric field amplitude ratio between different

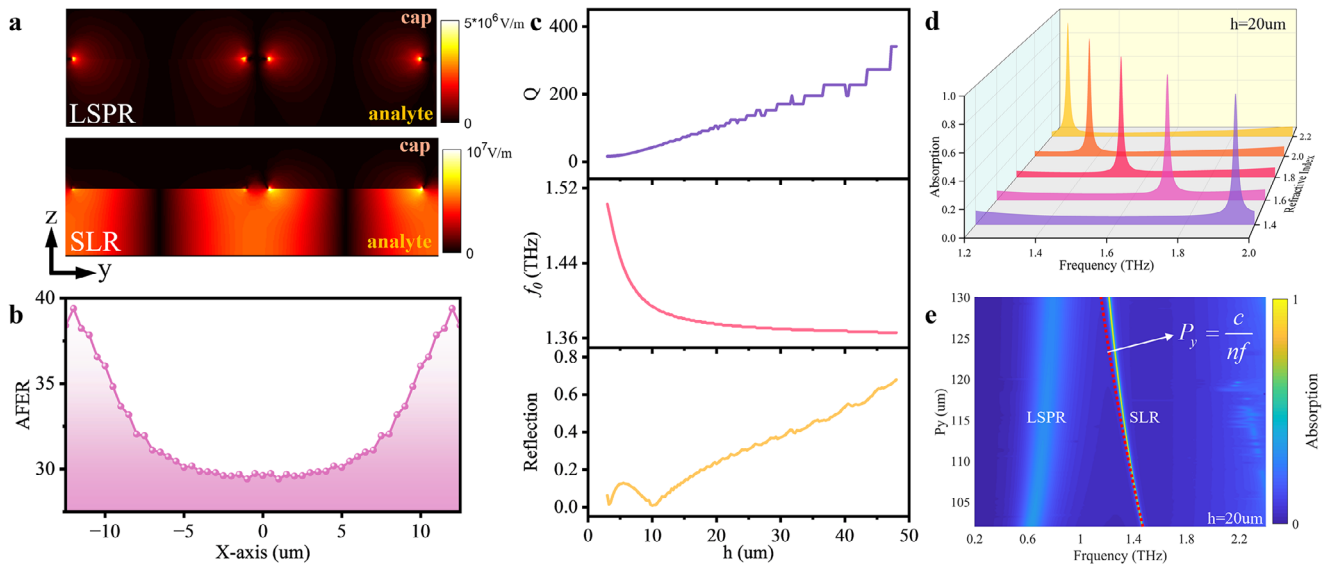


FIGURE 2 | Hybrid resonant mode analysis in LSW. (A) Lateral view of E-field distribution in LSW excited by LSPR and SLR respectively (unit: V/m). (B) Calculated AFER of LSW, where top metallic layer covers from -7 to 7 μm in X-axis. (C) Influence on Q-factor, resonant frequency f_0 and dip depth of LSW by varying channel height of LSW. (D) Absorption spectrum of LSW with different RI analyte with $h = 20$ μm . (E) The relationship between absorption spectrum and the period of lattice in Y direction of LSW with $h = 20$ μm .

resonant modes (SLR and LSPR) within the microfluidic channel as demonstrated in Equation (3), where V_{analyte} is the volume of channel. The calculated AFER of LSW is shown in Figure 2B, where field intensity of SLR is averagely 33 times greater than dipole resonance. Enhancement is particularly pronounced near the top metal layer edges, confirming the LSW's suitability for high-Q microfluidic resonators. Furthermore, AFER directly correlates with light-analyte overlap. Higher AFER indicates larger effective index variation of the waveguide mode [60], making AFER an effective metric for predicting resonator Q-factor and sensitivity.

$$\text{AFER} = \frac{\iiint_{V_{\text{analyte}}} \frac{E_{\text{SLR}}}{E_{\text{LSPR}}} dx dy dz}{V_{\text{analyte}}} \quad (3)$$

Next, we investigate the microfluidic channel height in the LSW design. Although open-boundary MIM resonators are promising for microfluidics due to their inherent compatibility with analyte flow, their performance is critically dependent on channel geometry. The channel height, a key yet understudied parameter in MIM structures, is typically minimized to maintain a high Q-factor. This is because, according to MIM theory, the radiation loss of the LSPR mode is inversely proportional to the dielectric layer thickness [61]. Therefore, achieving high Q-factors in conventional microfluidic integrated MIM resonators requires precise control of the channel height in terahertz band (often below 10 μm) [18, 27], shifting the challenge to the tolerances of micro/nano-fabrication technology. This imposes significant fragility on the sensing system, as even minor deviations in channel height can severely degrade the resonance. Consequently, resonant modes within MIM structures that tolerate or even benefit from increased channel heights are rare and highly desirable for robust metasurface-based sensors.

Here, to enhance the Q-factor, we utilize a waveguide mode excited by SLR as an alternative to the LSPR mode. When the wave vector is matched, the SLR structure excites a grating field that supports GMR of the slot waveguide. As shown in Figure S1, the collective resonance of SLR redistributes the field of dipole array and effectively quenching radiative leakage from interparticle gaps (See Section S1). Moreover, this approach transforms the problem from relying on surface plasmon resonance to leveraging a 3D waveguide mode, provided the mode coupling coefficient is sufficient. Consequently, the field confinement requirement for this open-sided MIM resonator is significantly relaxed, approaching that of conventional waveguides, as quantified by the field confinement factor (Γ):

$$\Gamma = \frac{\iint_{D_{\text{slot}}} \text{Re}\{E \times H^*\} e_z dx dy}{\iint_{D_{\text{tot}}} \text{Re}\{E \times H^*\} e_z dx dy} \quad (4)$$

where D_{slot} and D_{tot} represent the domain of slot and total region respectively. As described in Equation (4), a higher ratio Γ indicates greater concentration of electromagnetic energy. Calculation based on Equation (4) reveal that the effective index (n_{eff}) of cap layer governs the FWHM of Γ , which is another equivalent description of Q-factor. Furthermore, the SLR inherently exhibits a high n_{eff} , making it well-suited for exciting SW modes. While the SW supports multiple resonant modes, we select a high Q-factor transverse magnetic (TM) mode within our LSW design. Consequently, this LSW configuration inherits the high-Q characteristics of the TM mode and exhibits significantly reduced dependence on minimizing the channel height to achieve radiative suppression. This effectively mitigates the most significant inherent limitation of conventional MIM-based microfluidic sensors. (See Section S2)

The impact of channel height is analyzed in detail as shown in Figure 2C. Unlike typical MIM resonators, the calculated Q-factor of LSW is no longer dominated by radiation leakage from

the open boundary, instead exhibiting characteristics intrinsic to waveguide modes. Here, we utilize the parallel plate waveguide (PPWG) configuration as the analytical model since LSW shares two fundamental characteristics with a PPWG: both structures support transverse magnetic (TM) modes and feature laterally open boundaries. Therefore, the resonant frequency (f_0) of the LSW is determined by the propagation constant of the TM mode in PPWG, as described by Equation 5 [62, 63].

$$\beta_{TM} = \sqrt{\omega^2 \mu \epsilon - (m\pi/h)^2} \quad (5)$$

Given that the SLR propagation constant is fixed for a given lattice period, the resonant frequency (f_0) asymptotically approaches a fixed value as the channel height increases. Simulation results in Figure 2C align with this theoretical prediction, showing f_0 converging at 1.36 THz. However, larger channel heights introduce additional propagation loss in the SW [49]. Consequently, the absorption characteristics of the LSW are modified, resulting in progressive shallowing of the absorption dip depth.

Since the LSW is established on MIM configuration, CMT can be applied to analyze its characteristics [64]. In CMT framework, resonator losses are categorized as either absorptive or radiative, with corresponding quality factors defined as $Q_a = \omega_0 \tau_a / 2$ for absorption and $Q_r = \omega_0 \tau_r / 2$ for radiation. Our analysis reveals that the extracted damping ratios of $1/\tau_a$ and $1/\tau_r$ decrease significantly with increasing channel height. This dual reduction explains the enhanced total Q-factor ($Q_{total} = (1/Q_a + 1/Q_r)^{-1}$) observed in LSW across varying channel heights. However, this height-dependent Q-factor improvement occurs at the expense of absorption performance. To quantify this trade-off, we introduce the Fractional Loss Deviation (FLD), defined as $(\tau_r - \tau_a) / (\tau_r + \tau_a)$, which directly correlates with absorption dip depth and provides a perfect prediction for absorption dip depth with absolute error less than 0.013. (See Section S3)

The reflectance spectrum of the proposed LSW was simulated using CST Microwave Studio, exhibiting a sharp absorption peak under Y-polarized terahertz illumination as shown in Figure 2D. For a 20 μm thick analyte layer, the structure demonstrates remarkable sensing performance with a quality factor of 130, a maximum sensitivity of 2.215 THz/RIU, and a FOM of 106.5, significantly surpassing conventional MIM-based microfluidic sensors. While the LSW mitigates stringent channel height constraints typical of MIM resonators, height selection remains critical for further performance enhancement. As detailed in Section S4 and shown in Figure S4B, increasing the channel height to 47 μm elevates the Q-factor beyond 300 and the FOM beyond 240, which is a substantial improvement over the typical FOM values (~ 10) of standard MIM microfluidics. Notably, increasing the channel height also reduces the coupling efficiency between the SLR and GMR modes, manifesting as a decrease in resonance absorption depth that can complicate spectral identification. This effect can be mitigated by employing detection systems with higher sensitivity to compensate for the shallower dips.

Figure 2E illustrates the calculated dependence of LSW resonant frequency on lattice period (P_y). The absorption dips corresponding to LSPR and SLR exhibit opposing trends with increasing P_y .

Moreover, the SLR frequency follows the theoretical prediction in Equation (1), confirming its high-Q collective resonant nature. A minor deviation between simulation and theory emerges at larger P_y values due to frequency dependent n_{eff} variations and associated effective refractive index approximations. These results demonstrate that the LSW inherits the tunable resonance characteristics of SLR, with frequency directly controlled via P_y . Importantly, parametric studies confirm that collective lattice resonance dominates the response, with individual resonator properties exhibiting negligible influence. Specifically, resonator dimensional parameters negligibly impact the resonant frequency but modulate the Q-factor through modifications of the metal filling ratio. A summary of the dimensional parameter selection criterion for the LSW is provided in Table S1 (Section S5), based on a comprehensive parameter study.

The RI sensitivity of LSW can be theoretically estimated based on SLR theory. Consistent with RA diffraction theory and mathematical derivation, SLR sensitivity in the terahertz regime increases at smaller lattice periods [43]. As an SLR derived structure, the proposed LSW retains this advantage and consequently benefits from reduced period lengths (P_y). FEM simulations establish the fundamental relationship between lattice period and refractive index sensitivity. The results confirm that the LSW sensitivity follows an inverse square relationship with lattice period ($S \propto P_y^{-2}$), demonstrating excellent agreement with SLR theoretical models as depicted in Figure S6. (See Section S6).

2.2 | Polarization Multiplexing Lattice Slot Waveguide (PMLSW)

Our analysis establishes that the LSW leverages planar metallic structures to enhance field confinement beyond conventional MIM resonators. This capability enables highly feasible multiplexing, which is a critical advancement toward multifunctional terahertz microfluidic sensors. Unlike traditional MIM approaches, where polarization multiplexing suffers from intrinsic radiative losses, the collective resonance nature of LSW provides a fundamental solution. By implementing a rectangular lattice with C_2 symmetry (Figure 3A), orthogonal LSWs achieve polarization multiplexing with minimal cross-talk. Given that each meta-atom of LSW comprises two sub-units, we transform the lattice geometry and construct a composite meta-atom with four sub-units through cross multiplication of lattice coordinates ($\bar{Y} \times \bar{X}$). This configuration minimizes mutual coupling while maintaining resonance integrity, where patterned orthogonal lattices coexist in-plane with optimized isolation. A detailed design rationale for can be found in Section S10. The final implementation of PMLSW after optimization is illustrated in Figure 3A.

The PMLSW unit cell, developed through coordinate transformation (Figure 3B), retains the MIM sandwich structure of the original LSW. Implemented at 20 μm channel height, its simulated reflectance spectrum (Figure 3C,D) reveals dual-polarization resonances: under Y-polarization, distinct absorption peaks span 1.4 – 2.0 THz across analyte RI from 1.4 to 2.2, while X-polarization yields absorption at 2.0 – 2.6 THz. To enable multiplexing capabilities, the PMLSW exhibits a moderately reduced quality factor compared to the LSW, resulting from

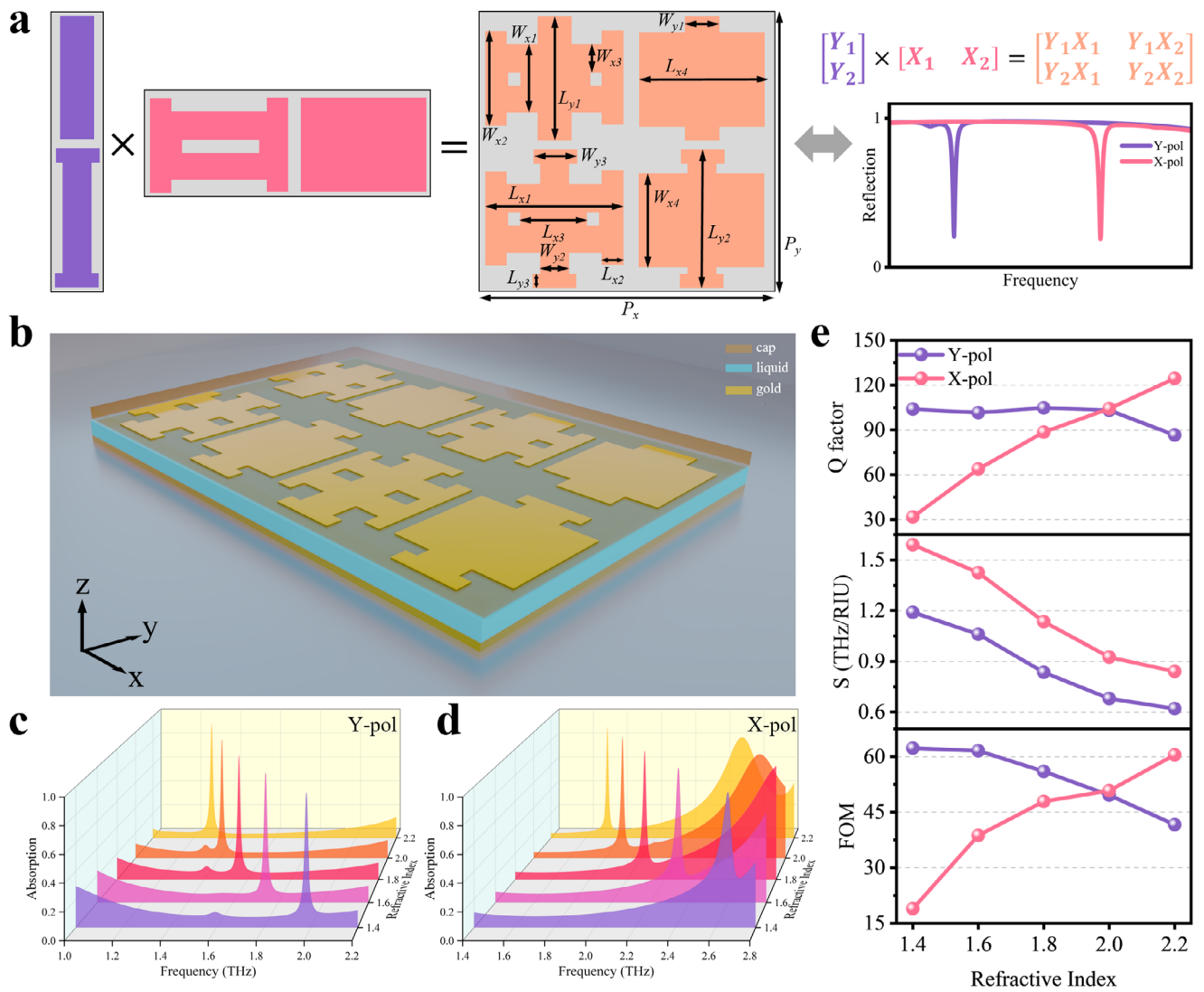


FIGURE 3 | Polarization Multiplexing of LSW. (A) Implementation of 2-D rectangular lattice with C_2 symmetry. (B) Unit configuration of PMLSW. Here $P_y = 110 \mu\text{m}$, $P_x = 80 \mu\text{m}$, $L_{y1} = 43 \mu\text{m}$, $L_{y2} = 49 \mu\text{m}$, $L_{y3} = 6 \mu\text{m}$, $L_{x1} = 38 \mu\text{m}$, $L_{x2} = 5 \mu\text{m}$, $L_{x3} = 13 \mu\text{m}$, $L_{x4} = 34 \mu\text{m}$, $W_{y1} = 16 \mu\text{m}$, $W_{y2} = 13 \mu\text{m}$, $W_{y3} = 22 \mu\text{m}$, $W_{x1} = 23 \mu\text{m}$, $W_{x2} = 32 \mu\text{m}$, $W_{x3} = 9 \mu\text{m}$, $W_{x4} = 30 \mu\text{m}$. (C) Reflective absorption spectrum of PMLSW in Y-polarization. (D) Reflective absorption spectrum of PMLSW in X-polarization. (E) Sensing performance of PMLSW with 20 μm channel height.

orthogonal period constraints that limit the value of metal filling ratio. However, PMLSW still achieves a high Q of more than 100 in both polarizations (Figure 3E), surpassing all comparable microfluidic chips in terahertz band. This performance stems from collective lattice resonance transforming surface waves to guided modes, suppressing mutual coupling significantly relative to conventional polarization multiplexed LSPR platforms. Sensitivity is fundamentally governed by polarization and can be tuned via the corresponding lattice period, where reducing P_x enhances sensitivity under X-polarization. Consequently, PMLSW achieves enhanced dual-polarized FOM, representing a significant advancement for MIM resonator based microfluidic sensors.

Consistent with LSW behavior, PMLSW performance is predominantly governed by lattice periodicity and channel height. Here we put forward a detailed study as illustrated in Figure 4. The impact of lattice period in the resonant frequency is studied in Figure 4A–D. According to LSW theory, the reso-

nance in microfluidics can be simply divided into two orthogonal modes categorized by polarized direction. Both LSPR and SLR in PMLSW are influenced only by the period in corresponding directions, while period length in orthogonal direction hardly affects. This demonstrates excellent isolation between orthogonal polarizations, with negligible mutual coupling.

Furthermore, a synthesis analysis of lattice periodicity on the dip frequency and Q-factor of SLR is demonstrated in Figure 4E–H. From Figure 4E,F, it is observed that the dip frequency is sensitive to the period in the corresponding polarization direction only, which confirms the effective of lattice decoupling. Moreover, Q-factor optimization is governed by the metal filling ratio, which highly correlates with the lattice period in orthogonal direction. As demonstrated in Figure 4G,H, enhancing P_y or P_x would influence the resonance strength in both polarization modes, and a more compact lattice is favorable for applications with higher Q-factor and sensitivity demands.

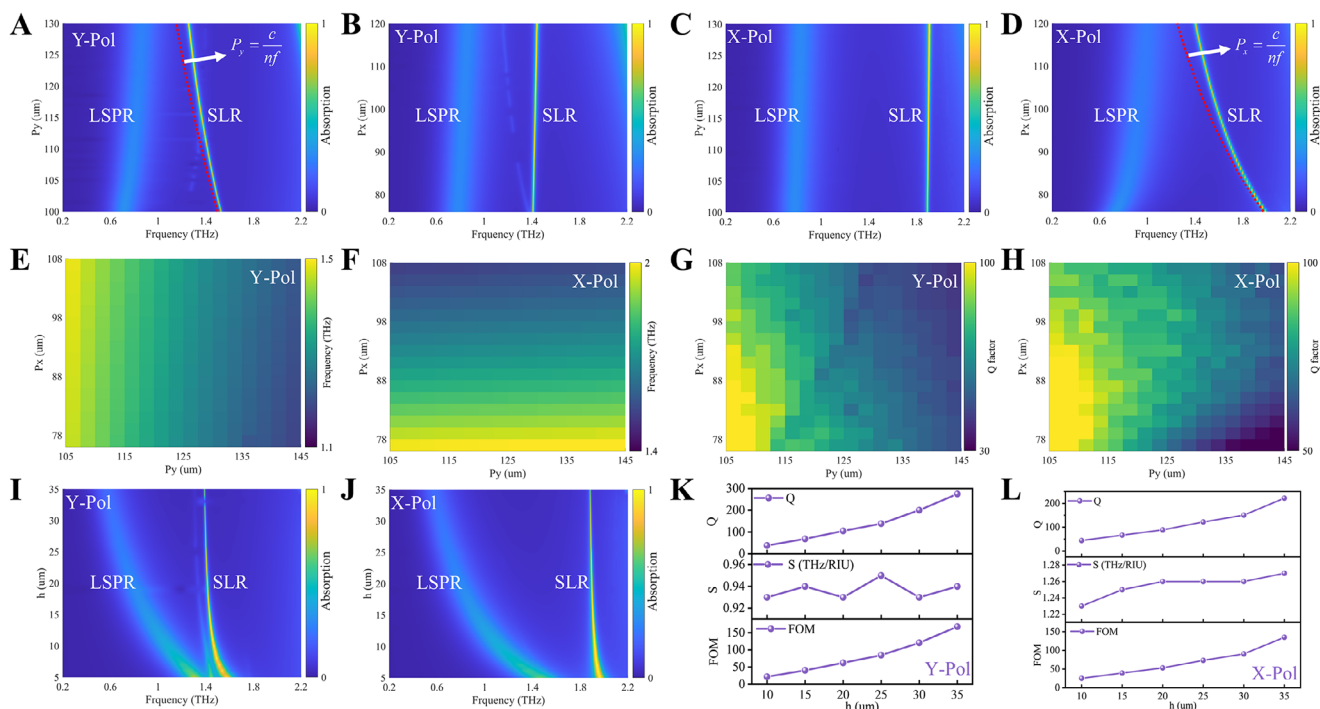


FIGURE 4 | Performance of PMLSW in orthogonal resonant modes. (A–D) Absorption spectrum of PMLSW with various P_y and P_x . (E–H) Q-factor and dip position of dual-polarized SLR with various periods in PMLSW. (I–L) Summary of the channel height impact on PMLSW.

Channel height critically modulates PMLSW performance as systematically investigated in Figure 4I–L. Unlike sensitivity controlled by period, channel height impacts Y-polarized and X-polarized modes simultaneously while preserving polarization isolation. As a leakage radiative mode, the LSPR resonant frequency exhibits significant dependence on the channel height, whereas SLR is relatively insensitive as illustrated in Figure 4I,J. A summary of sensing performance with different channel heights is listed in Figure 4K,L. The Q-factors of both modes enhance as the increase of channel height, which agrees well with the properties of LSW and a maximum Q factor of 275 can be observed at a channel height of 35 μm . On the other hand, the sensitivity of both modes remains steady as the period in X and Y directions remains unchanged, which can be estimated by P_x^{-2} and P_y^{-2} respectively. In general, the FOM of sensor would increase as channel height contributes to Q-factor only and has little influence on sensitivity. This brings extra advantages to PMLSW and a maximum FOM of 135 can be achieved in both polarization modes with 35 μm channel height.

To better demonstrate the lattice characteristic of PMLSW, the E-field distribution is depicted in Figure 5A,B. The polarization isolated lattice field distribution can be observed in LSW when exposed to Y-polarized and X-polarized THz wave respectively. The two collective resonance modes cover the whole interface and couples to GMR inside the channel. Based on field distribution information of PMLSW, the calculated results of AFRR are presented in Figure 5C,D. Similar to LSW, the AFER of PMLSW reaches peaks at the blank area without metallic coverage. A maximum field enhancement of more than 20 times can be observed in both polarization directions. This verifies the 2D patterning of SLR for polarization multiplexing and its capacity in coupling to SW in orthogonal directions. Therefore, this

architecture inherently suppresses mutual coupling to -50 dB while enabling dual-polarization sensing (See Section S10), which is a fundamental advancement for side-open microfluidics.

2.3 | Application of PMLSW in Biomedical Trace Analysis

Based on the analysis above, LSW architectures intrinsically enable high-Q resonance and precision RI sensing, particularly for biomacromolecule detection in aqueous environments. The slot inside LSW can be directly utilized as the channel of microfluidics. Therefore, a higher overlap degree between analyte and electromagnetic fields is realized because of a higher AFER. This engineered synergy makes LSW uniquely suited for trace biomolecular analysis in liquids. Here we experimentally validate this by sensing real analyte in aqueous environment with polarization multiplexing application.

The photography of fabricated prototype is presented in Figure 6A–F. It is a multilayer structure composed of a cap layer and a substrate layer, where metallic structures are deposited on PI and silicon by physical vapor deposition (PVD). The channel with 20 μm height is formed on silicon wafer by deep reactive ion etching (DRIE) method. After preparation, the whole chip is assembled by a homemade acrylic fixture with inlet/outlet ports (see details in Processing and assembly). To verify the Q-factor of LSW, no solution is injected into the microfluidics at first. The performance of this metasensor is characterized by THz-TDS system (QT-TRS1000, Qingdao Quenda Terahertz Technology Co., Ltd.). Normal incidence reflection lens is utilized to construct normal reflection measurement and the spot position is aligned based on the reflective imaging system with

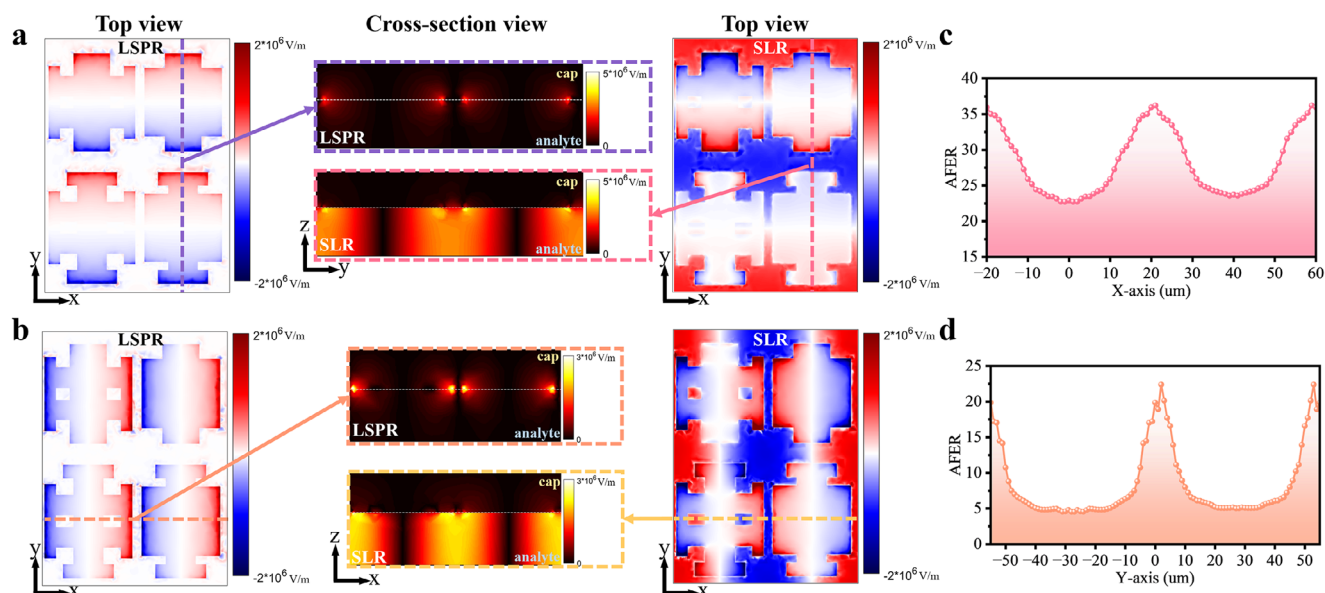


FIGURE 5 | Mode analysis of PMLSW. (A) E-field distribution in PMLSW under illumination of Y polarized wave. (B) E-field distribution in PMLSW under illumination of X polarized wave. (C) Calculated AFER of PMLSW under illumination of Y-polarized wave. (D) Calculated AFER of PMLSW under illumination of X-polarized wave.

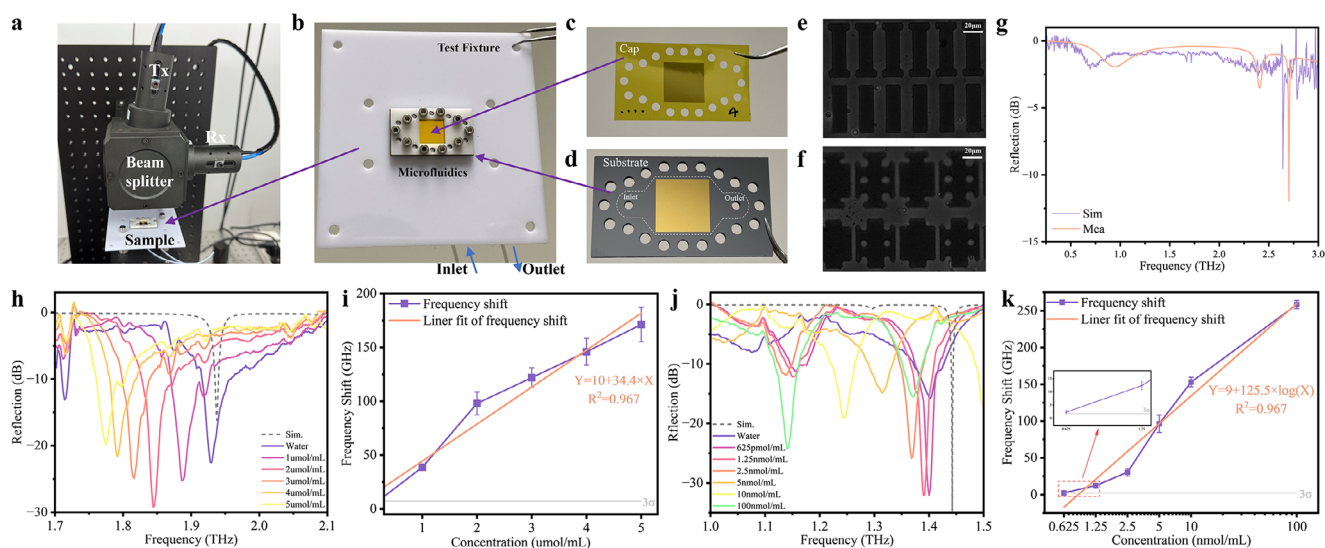


FIGURE 6 | Experimental verification of LSW. (A) Photograph of optical fiber THz time-domain spectroscopy with normal reflection lens. (B–D) Optical image of fabricated prototype, cap and substrate. (E,F) Microscopic image of fabricated cap layer. (G) Simulated and measured results of LSW. (H,I) Simulated ($n_{\text{water}} \approx 2$) and measured sensing results of glucose for X polarized PMLSW. (J,K) Simulated ($n_{\text{water}} \approx 2$) and measured sensing results of BSA for Y polarized PMLSW.

2-D electrically controlled displacement platform (See Section S8 and Figure 6A). The measurement results of unloaded LSW are demonstrated in Figure 6G, where the spectrum of LSPR and SLR exist at 0.72 and 2.642 THz respectively. The maximum Q-factor of our LSW is proved to reach 189 with air analyte.

Subsequently, we verified the sensing capability of PMLSW microfluidics. Here we prepare two solutions, Bovine Serum Albumin (BSA) solutions with concentration spanned 625 pmol mL⁻¹ to 100 nmol mL⁻¹ (100, 10, 5, 2.5, 1.25 nmol mL⁻¹, and 625 pmol mL⁻¹), while glucose concentrations ranged from 1 to 5 $\mu\text{mol mL}^{-1}$ (5, 4, 3, 2, 1 $\mu\text{mol mL}^{-1}$). Solutions

were infused via syringe injection, and the microfluidic design enabled efficient residue ejection by air injection, eliminating specialized cleaning and enhancing analysis throughput. Polarization multiplexing was employed, with glucose detection using X-polarization and BSA detection using Y-polarization. For glucose in X-polarization (Figure 6H,I), the resonant frequency shifted from 1.93 to 1.77 THz as concentration increased from 0 to 5 $\mu\text{mol mL}^{-1}$, yielding a sensitivity of 31.14 GHz/($\mu\text{mol/mL}$) ($R^2 = 0.967$) from linear fitting. For BSA in Y-polarization (Figure 6J), a 260 GHz resonance shift occurred at 100 nmol mL⁻¹ versus pure water. Experimentally, at a concentration of 625 pmol mL⁻¹, a frequency shift of 2 GHz

was observed with an uncertainty (σ) of 0.577 GHz (equivalent to 83 ng change in our microfluidics with channel volume of $1\text{cm} \times 1\text{cm} \times 20\mu\text{m}$). Using the 3σ criterion, the concentration at which the signal exceeds three times the noise (i.e., $2\text{GHz} > 3 \times 0.577\text{GHz}$) is defined, establishing an experimental limit of detection (LOD) of 625 pmol mL^{-1} . Given the molecular weight of BSA (66.446 kDa) and its significant impact on solution refractive index, sensitivity analysis used logarithmic concentration scaling (Figure 6K). Linear fitting demonstrated an average frequency shift of 126.6 GHz per decade of concentration change ($R^2 = 0.967$). In addition, the well-controlled frequency drift, along with repeated measurements, confirms the system's stability. The quantified frequency-shift uncertainty further validates a low experimental LOD of 625 pmol mL^{-1} , demonstrating the device's consistently high sensitivity (See Section S9). Furthermore, the polarization-multiplexing functionality was validated by simulations and experiments under diagonal (45°) incidence, which successfully excited the two orthogonal modes simultaneously, confirming the potential for dual-band sensing (See Section S10).

Experimental results demonstrate that our LSW significantly enhances the Q-factor of MIM resonators compared to existing microfluidic approaches (See Section S11). LSW excitation relaxes the stringent prior constraints on channel height, thereby substantially improving fabrication tolerance. Furthermore, LSW, directly excited by the metallic surface structure, is inherently well suited for multiplexing applications. Moreover, PMLSW exhibits negligible mode coupling effects, enabling independent high-performance sensing at both polarizations. This capability facilitates label-free, polarization-multiplexed detection of distinct analytes with high sensitivity in microfluidic platform for the first time. Collectively, this establishes a novel pathway to develop high-Q, multiplexable microfluidic THz chips incorporating side-open resonators, promising enhanced information capacity for THz absorption spectroscopy and significantly improved sensing efficiency.

3 | Discussion

Conventional THz trace analysis in microfluidics suffers from intrinsically limited Q-factors due to radiation losses through side-open channels. This inevitably constrains the sensing performance of microfluidics, particularly for trace detection in aqueous environment. To address this, we integrate metasurfaces with microfluidics via an LSW. This synergy provides greater design freedom and readily enables polarization multiplexing, realized here as a PMLSW. The PMLSW achieves a FOM exceeding 10^2 under dual polarizations, representing hundreds of times enhancement over existing polarization multiplexing microfluidics [65]. This advancement is extensible to other multiplexing or fingerprint recognition applications with high Q-factor demands. It is worth noting that the high Q-factor achieved through strong radiation loss suppression inherently increases the resonator's sensitivity to the absorption loss of the analyte. While this differs from conventional MIM sensors, it also presents a distinct design consideration: the trade-off between ultra-high Q and robustness against lossy media is an important axis for optimization in

future sensor designs. Further loss reduction can be pursued via experimental optimization, advanced low-loss materials, and resonator design enhancements such as increased metal filling ratio or added channel boundaries.

This platform demonstrates experimentally validated capability for detecting trace-level analytes like BSA and glucose in aqueous environments. Both resonance modes within the PMLSW exhibit exceptional responsivity to minute refractive-index changes, yielding a remarkable LOD of 625 pmol mL^{-1} . Since the resonant frequencies are primarily determined by the lattice period, the operational wavelength can be readily tuned. This inherent frequency tunability, combined with the built-in polarization multiplexing capability, substantially augments the data dimensionality at diagonal polarization. It facilitates precise analysis of multi-component mixtures and creates opportunities for unique interactions with anisotropic fluids, showing substantial promise for future high-accuracy and high-throughput sensing applications.

4 | Materials and Methods

4.1 | Numerical Calculations

Numerical simulations of LSW and polarization multiplexing characteristics were performed using the frequency domain solver in CST Microwave Studio based on the FEM. Periodic boundary conditions were applied in x and y directions, with open boundary conditions in z direction. First two modes of Floquet ports are utilized corresponding to liner polarized wave in y and x directions respectively. Geometric parameters were optimized to achieve a balanced trade-off between Q-factor, sensitivity, and resonance dip depth. While lattice period primarily determines resonant frequency, Q-factor and sensitivity exhibit stronger dependence on unit cell structural parameters. Material's relative permittivity in the THz range was modeled according to Drude model in CST. Analyte refractive indices were modeled without intrinsic losses in simulations.

4.2 | Processing and Assembly

The microfluidic chips were fabricated using standard lithography processes. First, a $15\text{-}\mu\text{m}$ thick PI film was spin-coated onto a carrier substrate and thermally cured under nitrogen atmosphere. A 10-nm chromium adhesion layer and 200-nm gold layer were subsequently deposited via sputtering. Photolithography with positive photoresist and lift-off processing defined the metasurface lattice on the PI substrate. Second, a 1-mm silicon wafer was etched by deep reactive ion etching (DRIE) to form microfluidic channels, followed by sputter deposition of an identical Cr/Au bilayer ($10\text{ nm}/200\text{ nm}$) within the channels to serve as the bottom reflector. Laser drilling created 2-mm diameter mounting holes in the silicon. Third, a laser-cut acrylic fixture incorporating fluidic inlets/outlets was prepared. Finally, the PI metasurface substrate and silicon microfluidic layer were aligned, mechanically secured within the acrylic fixture, which enables effective sealing while protecting the fragile silicon substrate by evenly distributing adjustable clamping pressure through its integrated silicone gaskets.

Acknowledgements

This study was supported by the Natural Science Foundation of Jiangsu Province for Youth (BK20240626), the Natural Science Research Start up Foundation of Recruiting Talents of Nanjing University of Posts and Telecommunications (NY222060), the Open Research Fund of State Key Laboratory of Millimeter Waves, Southeast University (No. K202534), the National Natural Science Foundation of China (62561014).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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